# Estimating Wetland Carbon Storage Change Google Earth Engine Web Application User Guide

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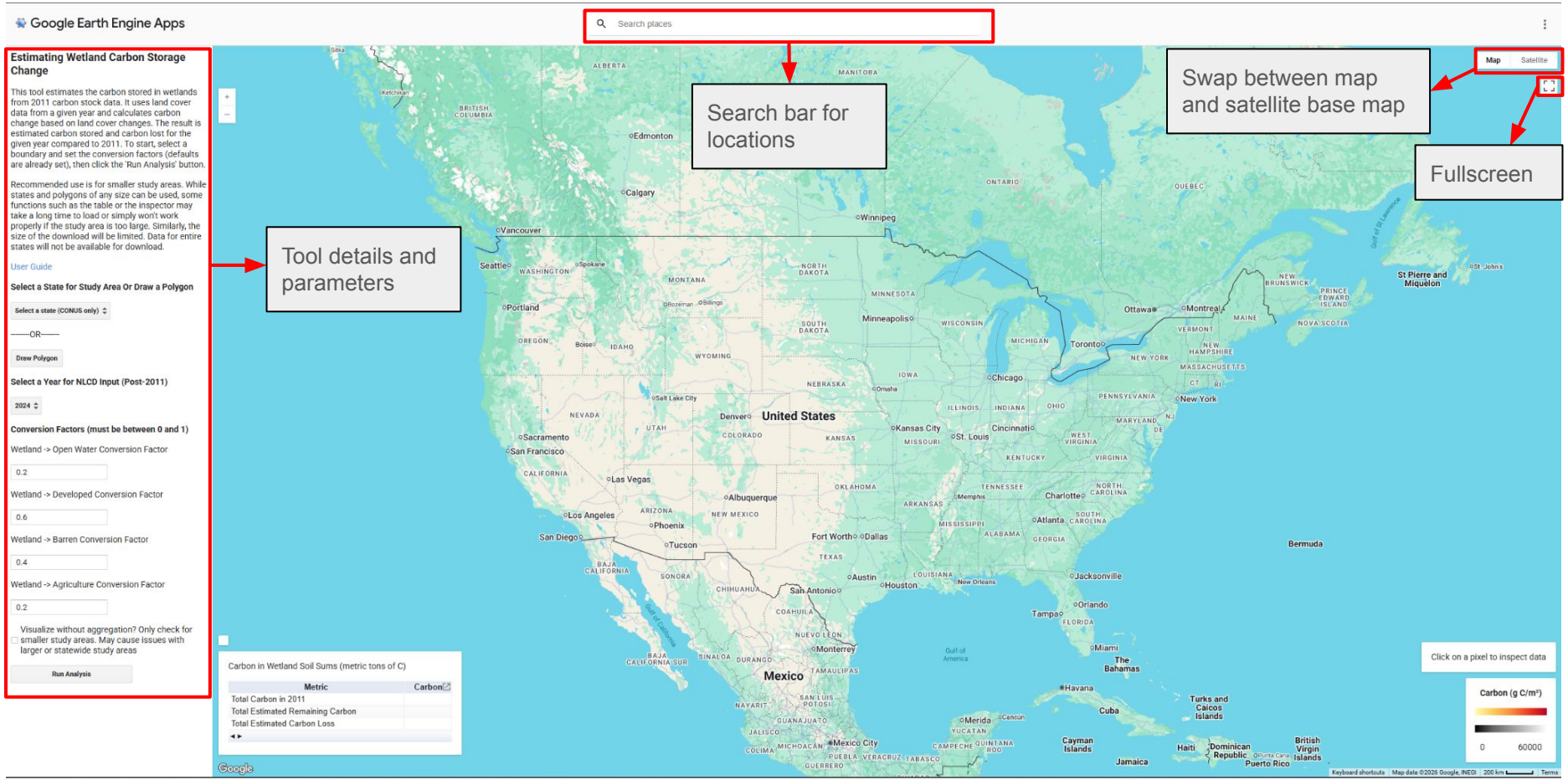
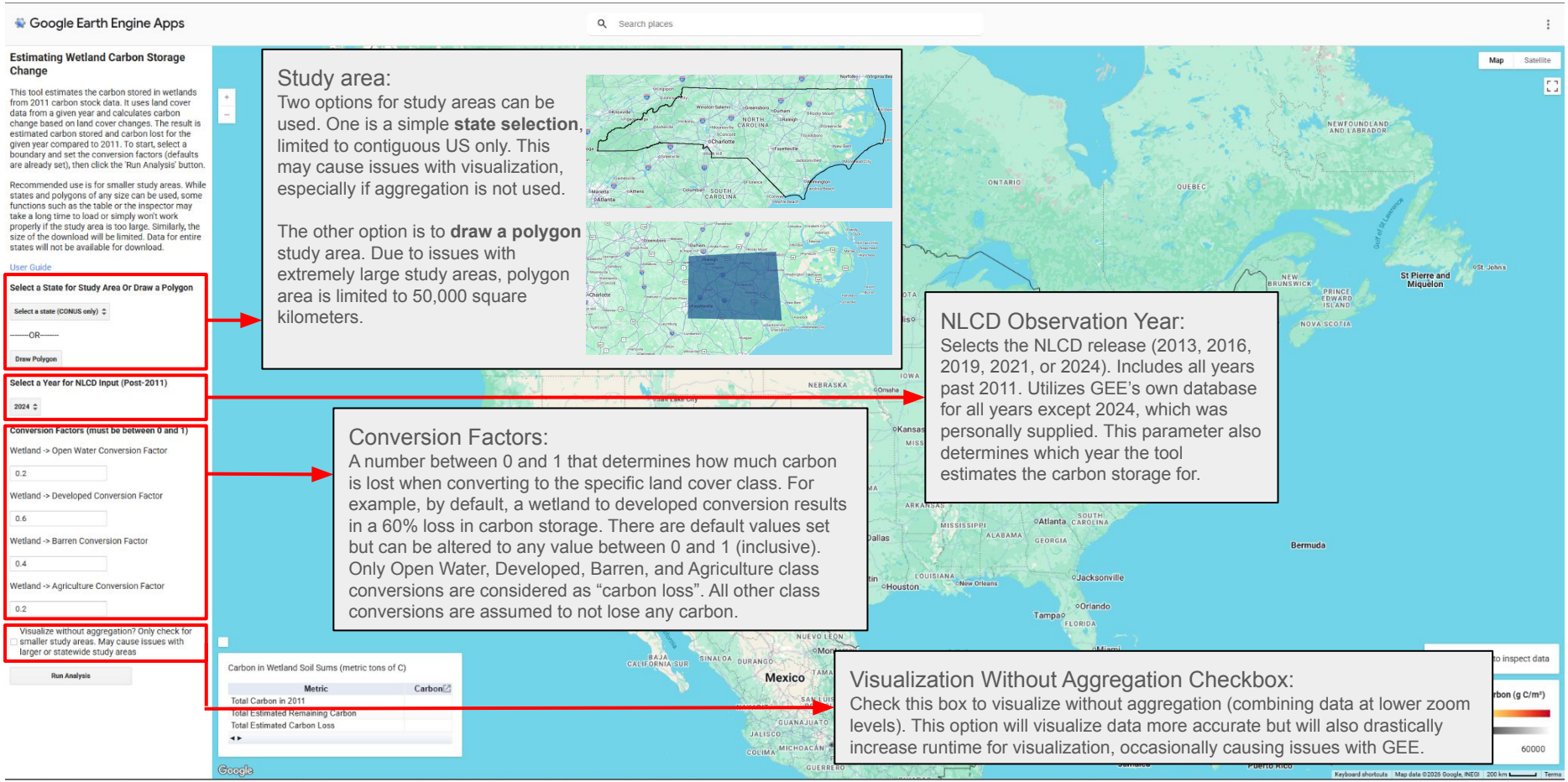
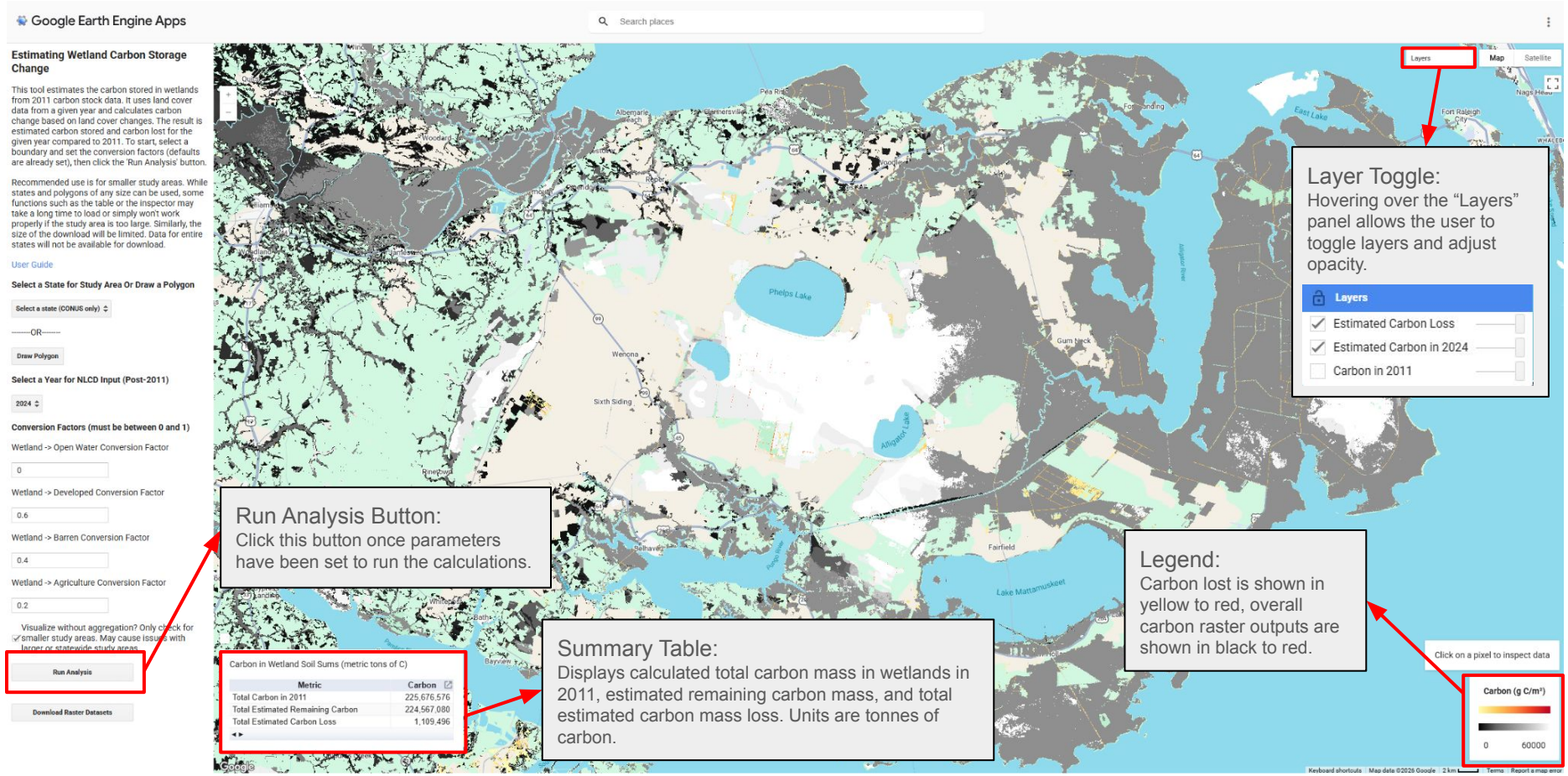
# Basic GEE Overview



# Tool Parameters



# Running and Results




## Layer Toggle:

Hovering over the "Layers" panel allows the user to toggle layers and adjust opacity.

## Run Analysis Button:

Click this button once parameters have been set to run the calculations.

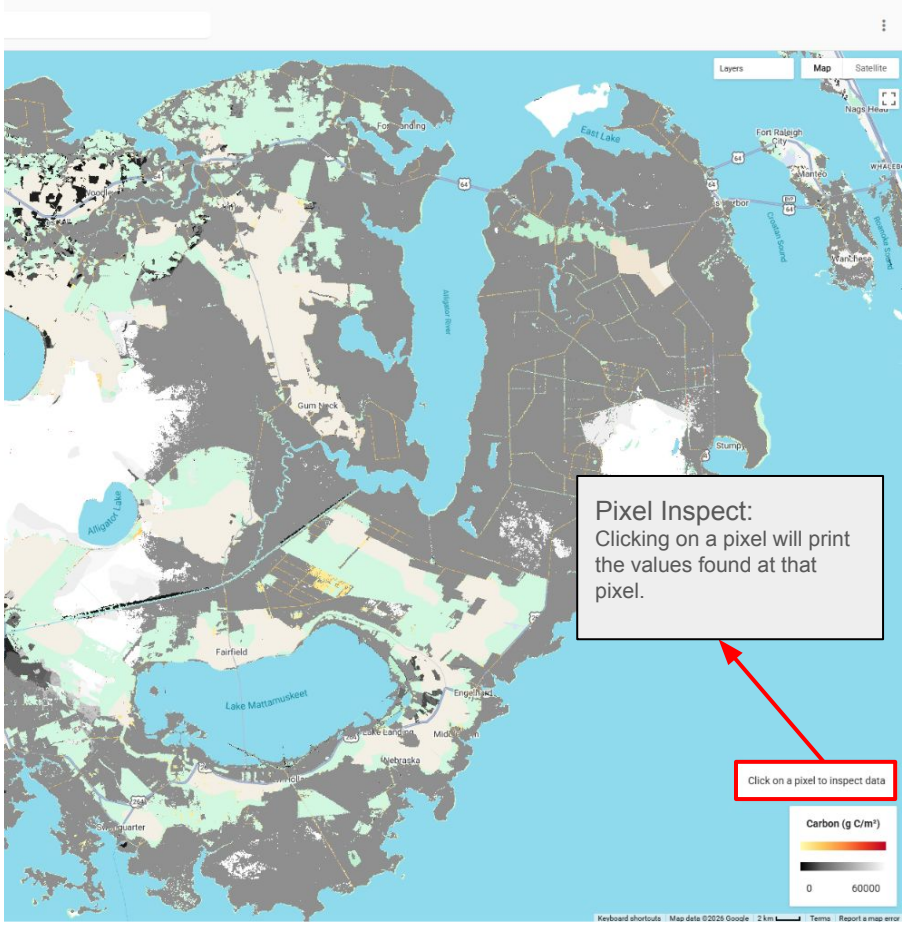
## Summary Table:

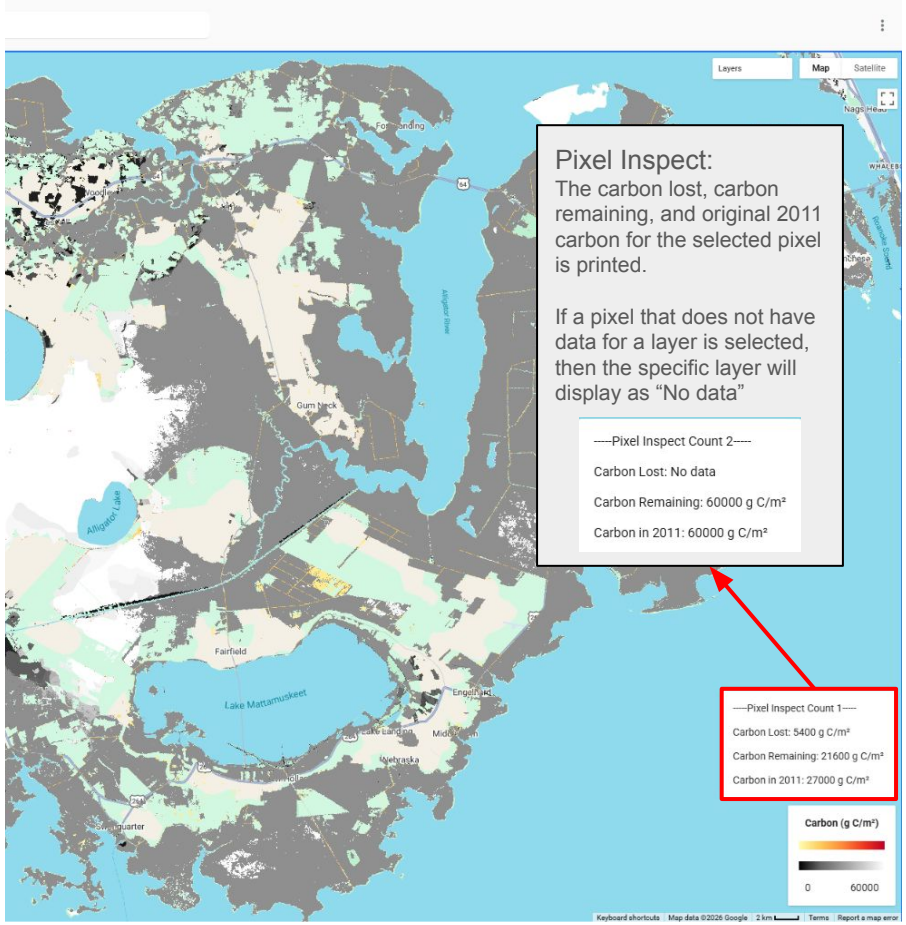
Displays calculated total carbon mass in wetlands in 2011, estimated remaining carbon mass, and total estimated carbon mass loss. Units are tonnes of carbon.

## Legend:

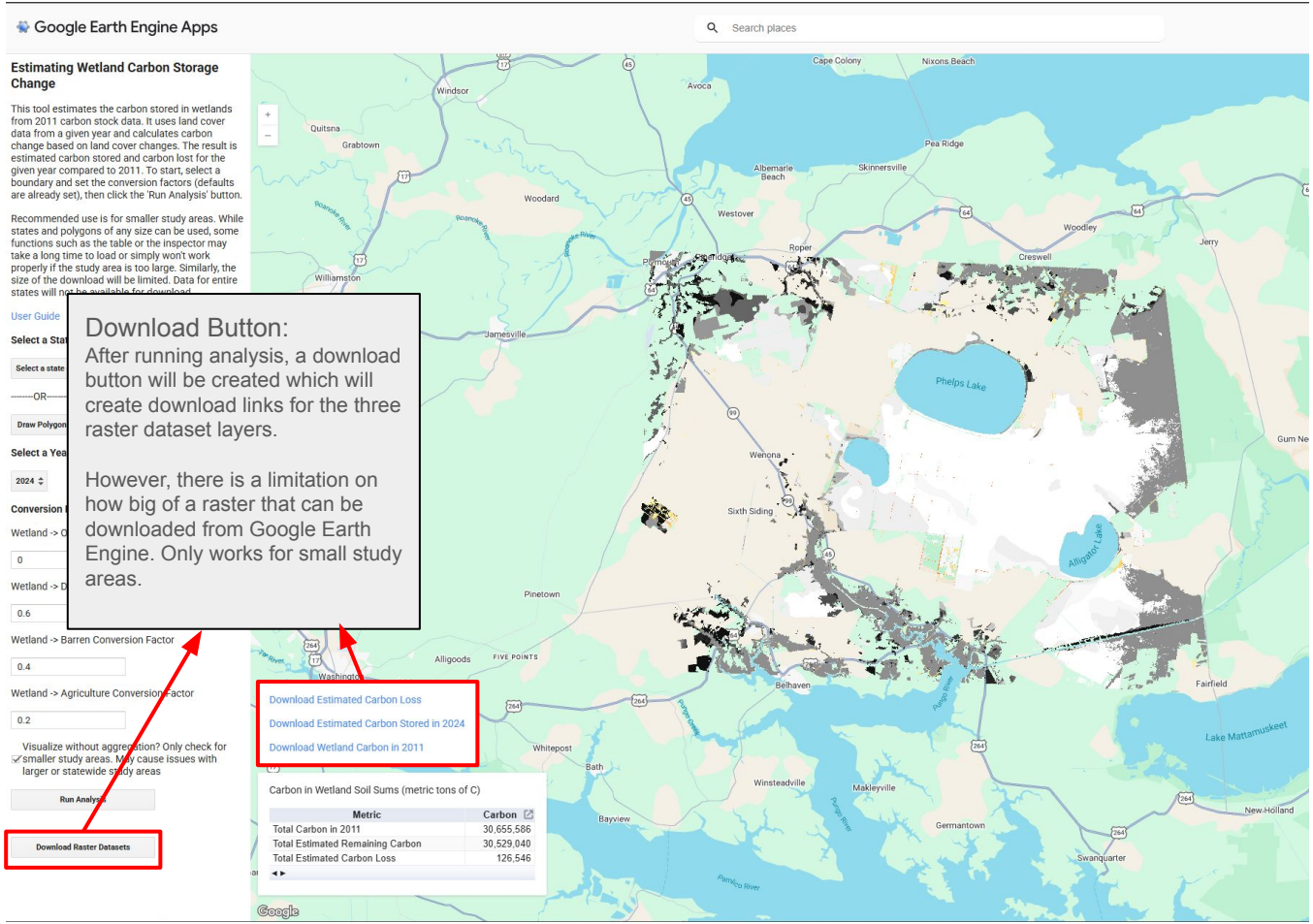
Carbon lost is shown in yellow to red, overall carbon raster outputs are shown in black to red.

# Pixel Inspect



# Downloading Raster Datasets



# Calculate Wetland Carbon Change ArcGIS Pro Custom Tool User Guide

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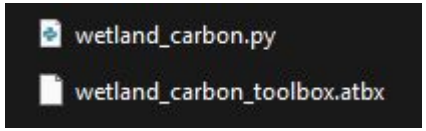
# Installation

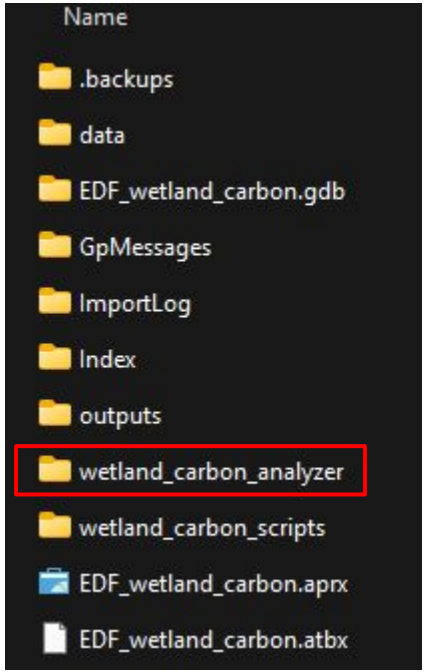
**Folder Installation:**

A zipped folder called *wetland_carbon_analyzer* will be provided with all necessary files. Inside will be documentation and this user guide. The two main files needed to run the ArcGIS Pro custom tool is the toolbox file and the Python script. Unzip this folder

Place the *wetland_carbon_analyzer* folder inside the project folder of an ArcGIS Pro project for easy use. This step is technically not necessary but will make the tool easy to access within ArcGIS Pro.

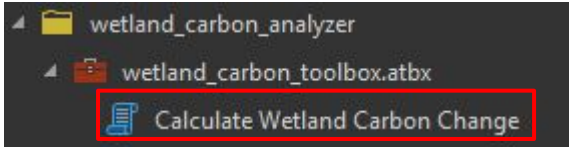
What is most important is that the ***wetland_carbon_toolbox.atbx*** **file and** ***wetland_carbon.py*** **Python script are placed in the same folder at the same level.** Otherwise, the custom tool will have to be manually linked to the Python script.



# Linking Python Script to Tool

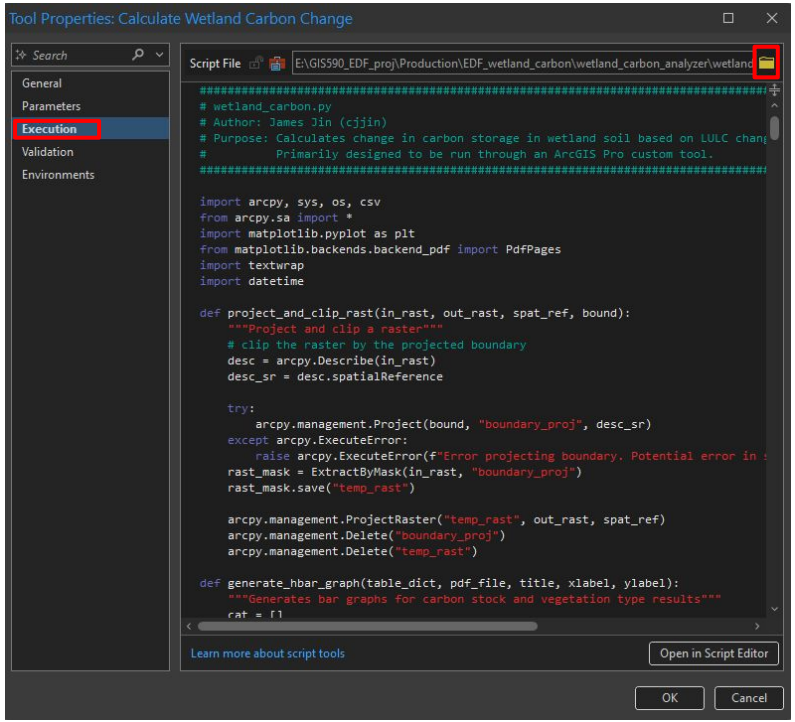
In the case that the tool has been disconnected from the script, then the user will have to manually connect it again.




Open ArcGIS Pro and navigate to the *wetland_carbon_toolbox* toolbox within the *wetland_carbon_analyzer* file. Expand it to find the *Calculate Wetland Carbon Change* tool. Right click it and click on **Properties.**

From the **Properties** window, click on the **Execution** tab. At the top of the window, where the script file pathos located, click on the folder icon at the top right.

Navigate to the *wetland_carbon.py* Python script file and select it. Then click OK. It should look like the image on the right with the script shown. Press OK to exit the **Properties** window. Now the tool should be linked to the script properly.



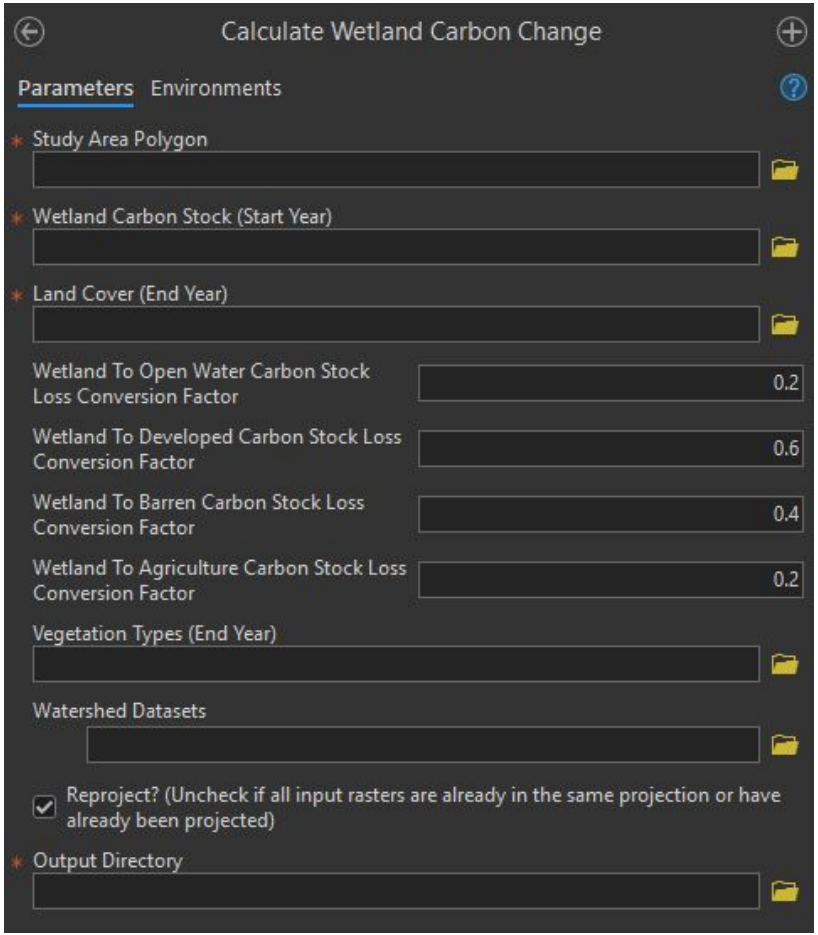
# Tool Overview

To open the tool, open ArcGIS Pro and navigate to the *wetland_carbon_toolbox.atbx* toolbox in *wetland_carbon_analyzer*. Double click on the *Calculate Wetland Carbon Change* tool to open the tool. The tool interface will look like the image on the right.

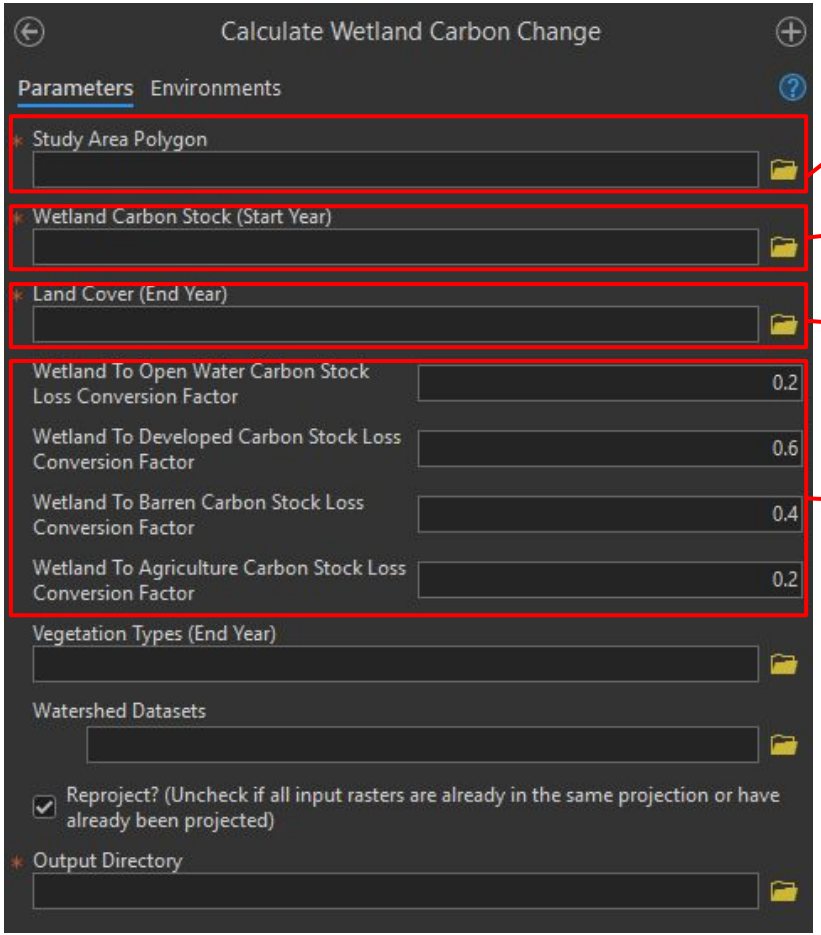
This tool uses a NLCD release and a 2011 Wetland Soil Carbon Storage dataset to calculate the estimated carbon storage in wetlands for the given NLCD release year.

[2011 Wetland Soil Carbon Storage Dataset Download]

[NLCD Download]

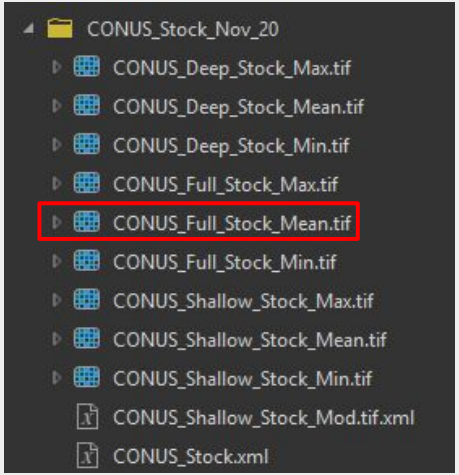


# Required Input Parameters




**Study Area Polygon:**
Defines the spatial extent of the analysis. Must be a polygon feature class/shapefile.

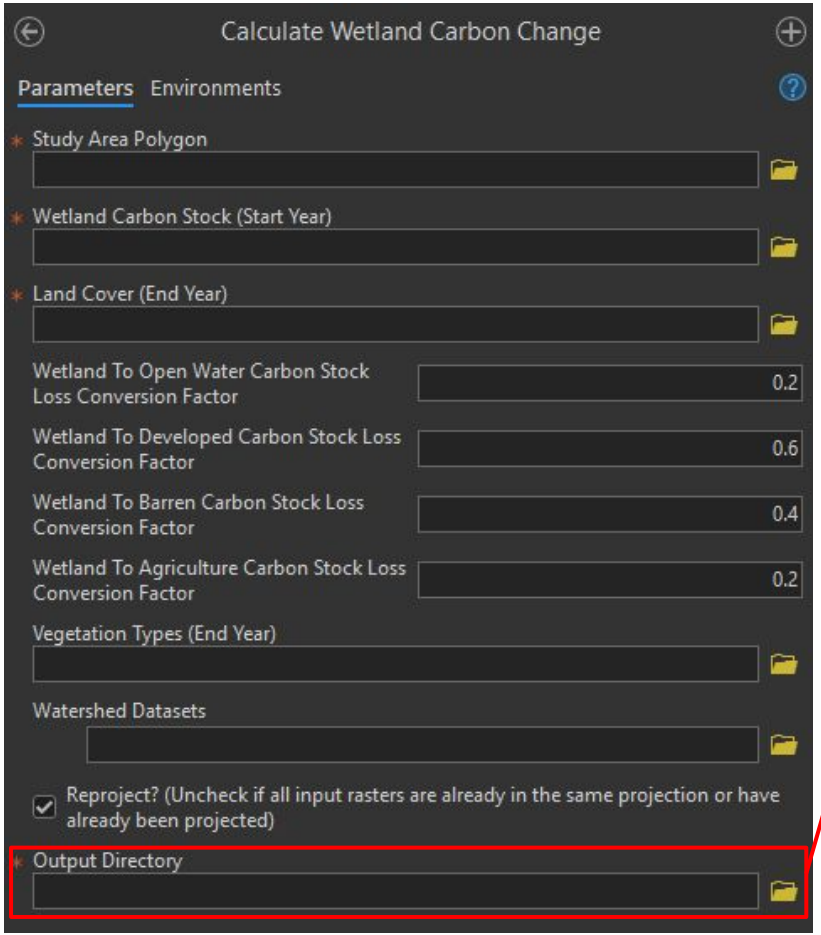
**Wetland Carbon Stock:**
Carbon stored in wetland soil data, this data is used as carbon data for the "before" time frame. The default is the CONUS full stock mean dataset but the other carbon stock data is valid.




**Land Cover:**
NLCD data used to determine conversion of wetland to other land classes. The year the NLCD release was in determines what year carbon storage is estimated for.

**Conversion Factors:**
A number between 0 and 1 that determines how much carbon is lost when converting to the specific land cover class. For example, by default, a wetland to developed conversion results in a 60% loss in carbon storage. There are default values set but can be altered to any value between 0 and 1 (inclusive). Only Open Water, Developed, Barren, and Agriculture class conversions are considered as "carbon loss". All other class conversions are assumed to not lose any carbon.

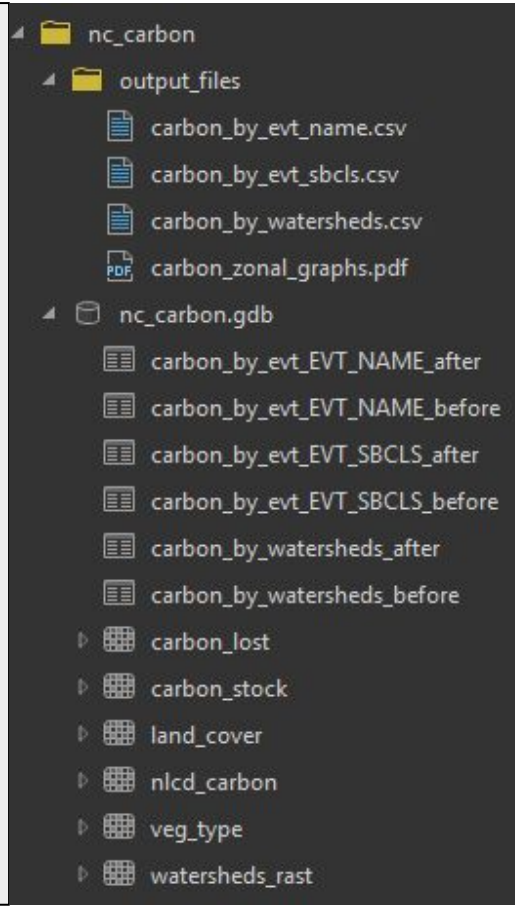
# Required Parameter: Output Directory



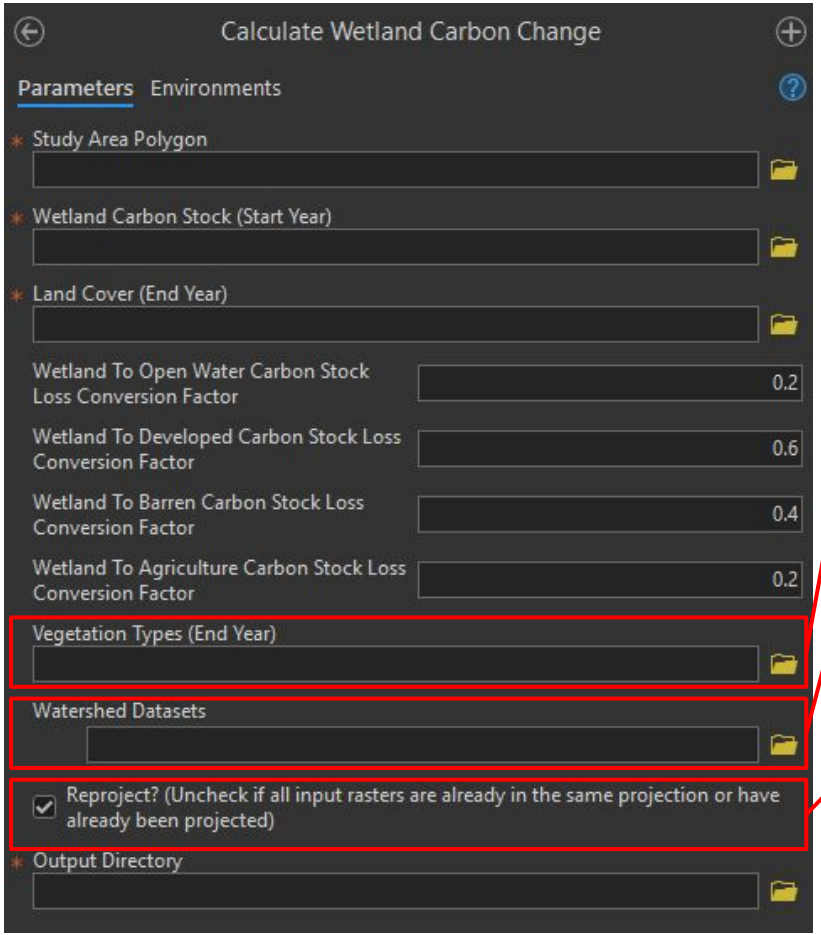

Output Directory:
Location where outputs and geodatabase will be created. Must be an existing directory, but does not need to be empty. However, outputs from previous runs will be overwritten if the directory and file names are kept the same.

An *output_files* folder and file geodatabase with the same name as the output directory will be created. The *output_files folder* contains all CSV and PDF output files.These files are only created if the optional parameters are also filled.

The file geodatabase will contain all outputs from geospatial processes. These raster datasets and tables will always be created. The file geodatabase will also serve as a workspace where intermediary files are stored and deleted after the tool finishes running. If for any reason the tool fails, it is recommended that the created file geodatabase is deleted to prevent crashing issues.



# Optional Parameters




## Vegetation Types:

Landfire Existing Vegetation Type data, used for zonal statistics. The year for Landfire EVT data should match the NLCD release year. If no input is read, then no vegetation type csv file will be created.

Download link for Landfire EVT data

## Watershed Datasets:

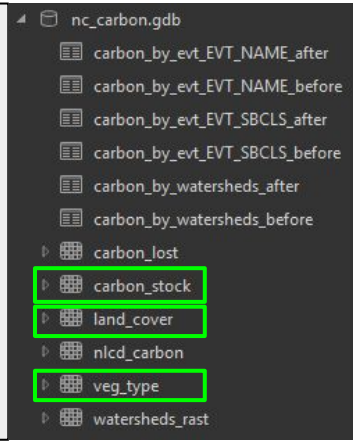
Watershed Boundary Dataset from USGS, used for zonal statistics by watershed. Allowed for multiple inputs if watershed data comes in multiple feature classes. If no input is read, then no watershed csv file will be created.

Download link for watersheds

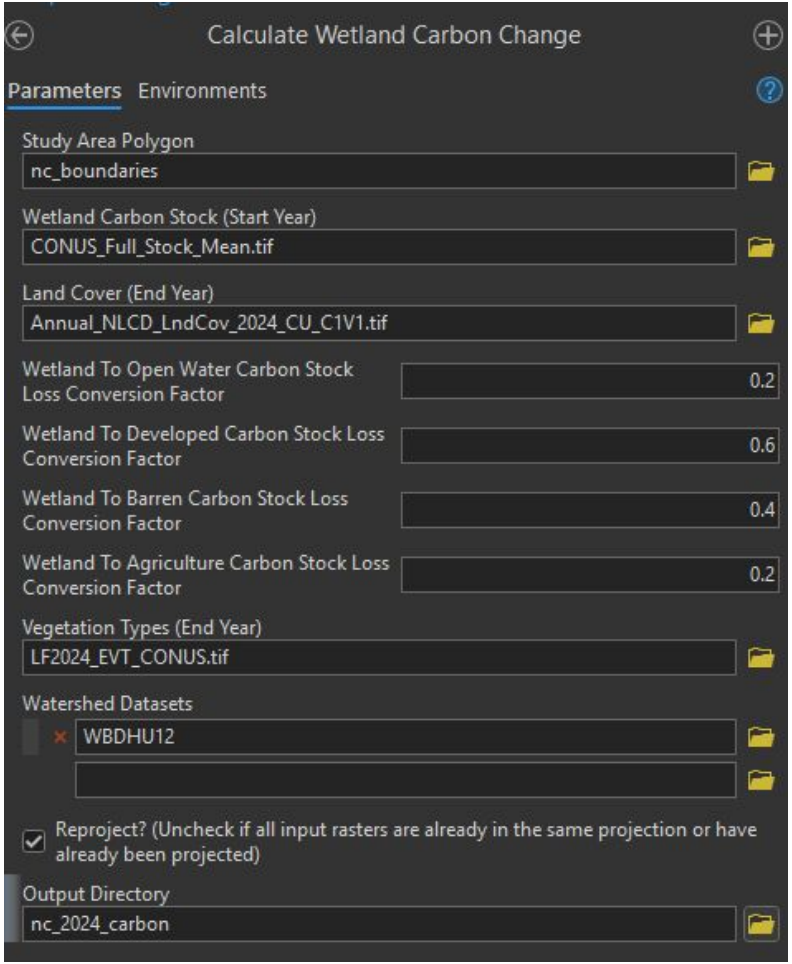
## Reproject and clip Checkbox:

Checking this checkbox (default state) enables projection (to NAD83 / Conus Albers) and clipping to the study area polygon. This is a step in the full process. However, if this step was already done and different parameters for conversion factors or watershed datasets were to be used, the projected and clipped raster datasets can be reused and the preprocessing can be skipped. Uncheck the checkbox to skip the projecting and clipping step.

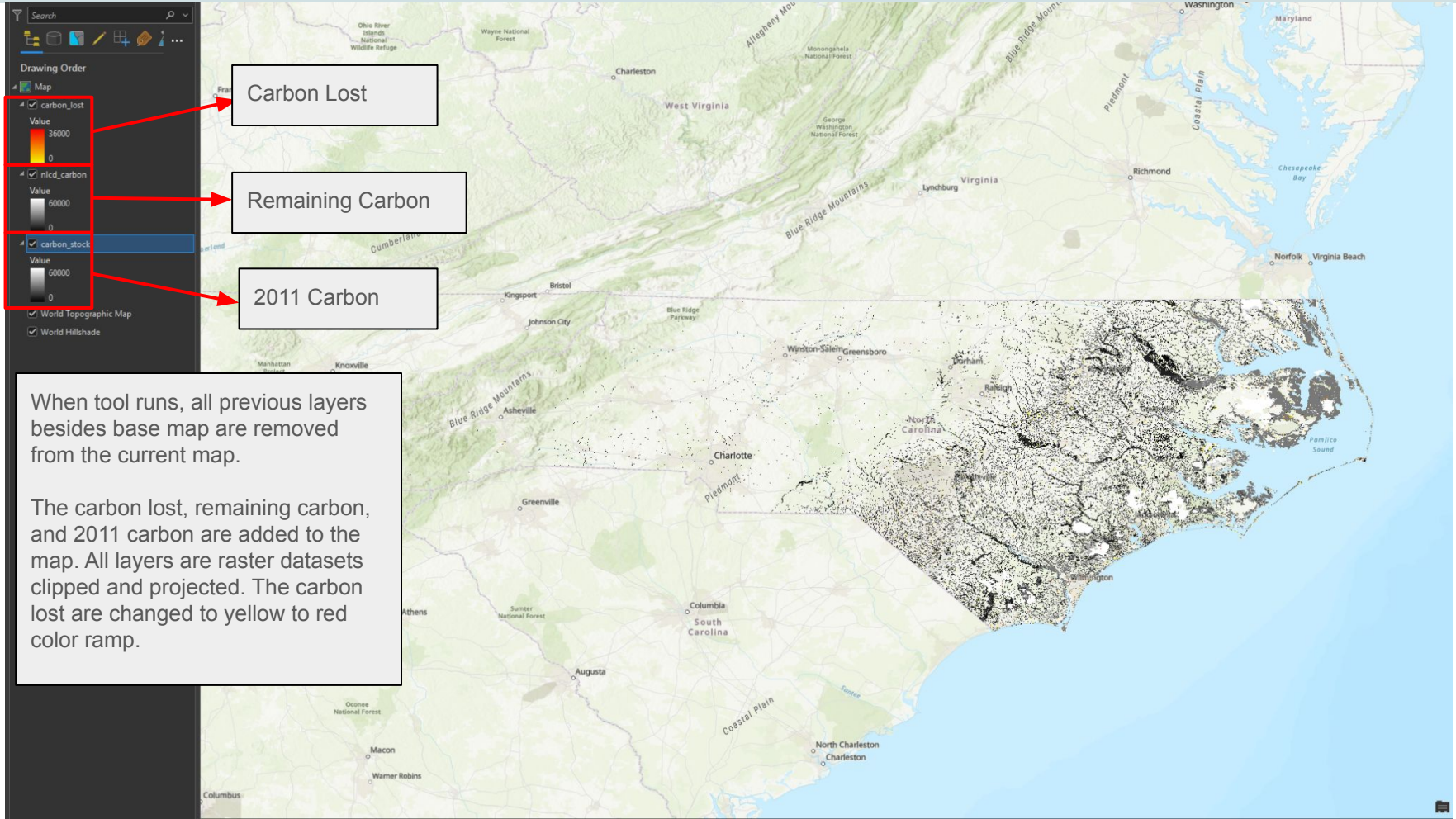
The raster datasets in green boxes are already projected and clipped from previous runs and can be reused.



# Example Input



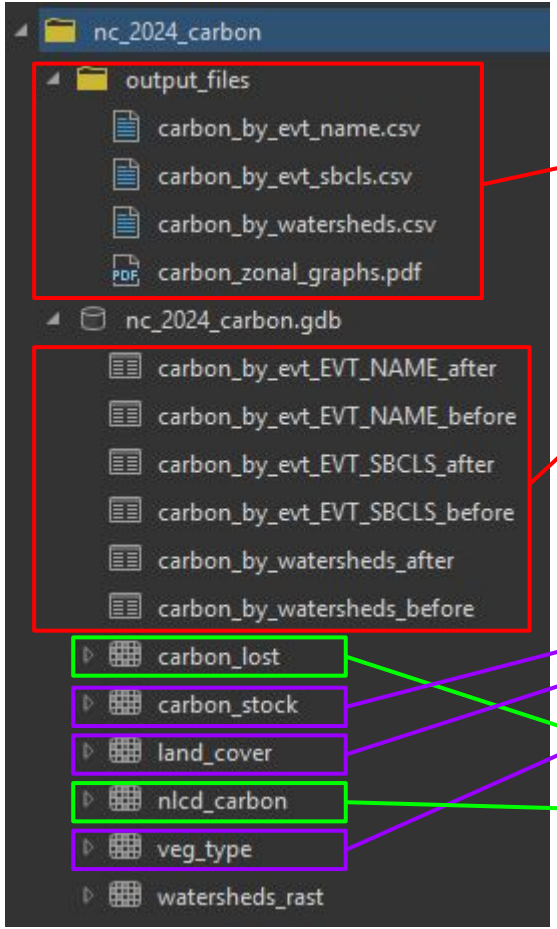
# Example Output




When tool runs, all previous layers besides base map are removed from the current map.

The carbon lost, remaining carbon, and 2011 carbon are added to the map. All layers are raster datasets clipped and projected. The carbon lost are changed to yellow to red color ramp.

# Output Files




## CSV and PDF files:

The CSV files are the results of zonal statistics calculations. Each CSV file displays carbon storage for 2011, NLCD release year, and carbon storage loss. The units are metric tons of carbon.

- *carbon_by_evt_name.csv* - zones usingLandfire EVT name (specific)
- *carbon_by_evt_sbcls.csv* - zones using Landfire EVT subclass (general)
- *carbon_by_watersheds.csv* - zones using watersheds

The PDF file only contains a bar graph for carbon storage loss for EVT subclass.

## Zonal Statistic Table Results:

Used for the CSV and PDF files.

## Projected and Clipped Input Raster Datasets:

These are the projected and clipped input raster datasets.

- *carbon_stock* - 2011 carbon storage
- *land_cover* - NLCD
- *veg_type* - Landfire EVT

These raster datasets can be reused as inputs for future runs when other parameters want to be changed.

## Actual Calculation Results:

These are the actual results for the wetland change calculations. The two raster datasets represent the estimated carbon/carbon lost in the NLCD release year.

- *carbon_lost* - estimated carbon lost
- *nlcd_carbon* - estimated remaining carbon

# Known Issues and Performance

This tool will take a few minutes to run. It can be long depending on study area and hardware on personal computer.

**IMPORTANT WARNING**

This tool is known to cause computer crashes on repeated runs. Closing ArcGIS Pro between runs may help.